MIL-C-55021/1A
31 October 1973
SUPERSEDING
MIL-C-55021/1
10 December 1965

MILITARY SPECIFICATION

CABLES: TWISTED PAIRS AND TRIPLES INTERNAL HOOKUP, SHIELDED AND JACKETED, MEDIUM TEMPERATURE

This specification is approved for use by all Departments and Agencies of the Department of Defense

1. SCOPE

1.1 Scope. This specification covers the detail requirements of synthetic resin insulated electrical hookup wire (jacketed, unjacketed, shielded, and unshielded) in twisted pair and triples for application in internal wiring of electric equipment at temperatures from -40 degrees Centigrade (C) to 105 degrees C (operating).

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Military

MIL-C-572	Cord, Yarns and Monofilaments Organic Synthetic Fiber
MIL-W-16878	Wire, Electrical, Insulated, High Temperature
MIL-M-20693	Molding Plastic, Polyamide (Nylon), Rigid
MIL-C-55021	Cables; Twisted Pairs and Triples, Internal Hookup, General Specification for

FSC 6145

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T22:16Z
Check the source to verify that this is the current version before use.

STANDARDS

Federal

FED-STD-228	Cable and Wire, Insulated, Methods of Testing

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement funcstions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Requirements. The complete requirements for procuring cable described herein shall consist of this specification and the issue in effect of MIL-C-55021.

3.2 Material. All material used in construction of cables furnished under this specification shall meet the requirements specified herein.

3.2.1 Insulated conductors. Insulated conductors used in construction of cables furnished under this specification shall conform to the requirements of MIL-W-16878, type B, C, or D, as specified (see 6.2 of MIL-C-55021).

3.3 Construction

3.3.1 Length of twist. The length of twist shall be as specified in Table I.

Table I - Length of Twist

Wire Size (AWG)	Twist (Inches)	Twist Tolerance (Inches)
32 to 24	2.0 (50.8)	±0.5 (12.7)
22 to 18	2.5 (63.5)	±0.5
16 to 14	3.5 (88.9)	±0.5
12 to 10	4.5 (114.3)	±0.5
8 to 6	5.5 (139.7)	±0.5
6 to 1/0	6.0 (152.4)	±0.5

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T22:16Z
Check the source to verify that this is the current version before use.

NOTE:

1. Dimensions are in inches.

2. Metric equivalents, to the nearest 0.1 millimeter (mm), are given for general information only and are based upon one (1) inch = 25.4 mm.

3.3.2 Covering over finished cable

3.3.2.1 Shield. When specified, a shield shall be applied over the wires. Shielding requirements shall be in compliance with MIL-W-16878.

3.3.2.2 Polyvinylchloride jacket thickness. When specified, a polyvinylchloride jacket shall be used. The thickness of the jacket shall be as specified in Table II.

Table II - Polyvinylchloride Jacket Thickness

Diameter Over Braided Shield	Jacket Thickness Nominal	Jacket Thickness Minimum
Inch	Inch	Inch
under 0.375	0.016	0.008
0.375 and over	0.025	0.015

3.3.2.3 Polyvinylchloride jacket physical requirements. The physical requirements of the polyvinylchloride jacket removed from the cable shall be as shown in Table III.

Table III - Physical Requirement

Property	Requirement
Tensile Strength (Pounds per Square Inch, Minimum)	1500
Elongation at Rupture (Minimum Percent)	100
Change in Elongation after Air Oven Aging at 100°C for 96 Hours (Percent of Unaged Elongation Maximum)	35

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T22:16Z
Check the source to verify that this is the current version before use.

3.3.2.4 Polyamide jacket. When specified, a clear, transparent jacket of heat stabilized polyamide, conforming to MIL-M-20693, comp A, type III, grade E, shall be used for OD beneath jacket of .250 inch or less. Over .250 inch beneath jacket a polyamide braid per MIL-C-572 shall be used. The thickness of the jacket shall be as specified in table IV.

Table IV Polyamide Jacket Thickness

Diameter Over Braided Shield	Jacket Thickness Nominal	Jacket Thickness Minimum
Inch	Inch	Inch
Under 0.200	0.006	0.004
0.200 to .250	0.008	0.006
0.251 and over	0.008	

3.3.2.5 Dimensional requirements for shield plus jacket. The maximum increase in diameter over the twisted multiple conductors resulting from the application of the braided shield and jacket shall be as specified in Table V.

Table V - Maximum Diameter Increase Due to Shield Plus Jacket

Diameter Over Shield	Diameter Increase Maximum (Polyvinylchloride)	Diameter Increase Maximum (Polyamide)
Inch	Inch	Inch
Under 0.375	0.057	0.051
0.375 and over	0.085	0.051

4. QUALITY ASSURANCE PROVISIONS

4.1 Destructive testing

4.1.1 Sampling. Sampling for destructive testing shall be in accordance with inspection level S-4 of MIL-STD-105. The AQL shall be 6.5 percent. Classification of defects shall be in accordance with table VI.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T22:16Z
Check the source to verify that this is the current version before use.

Table VI - Destructive Testing

Test	Requirement Paragraph	Test Paragraph	Classification of Defects Major	Minor
Tensile Strength	3.3.2.3	4.1.2	X	
Elongation	3.3.2.3	4.1.2	X	
Air Oven Test	3.3.2.3	4.1.2	X	

4.1.2 Physical properties (Polyvinylchloride). Polyvinylchloride jacket shall be tested in accordance with the following applicable test methods specified in FED-STD-228:

PROPERTY	TEST NO.
Tensile strength	3021
Elongation	3031
Elongation, after air oven aging (96 hours at 100°C)	4031

5. PREPARATION FOR DELIVERY

5.1 Preparation for delivery shall be in accordance with MIL-C-55021.

6. NOTES

6.1 Intended use. Cables covered by this specification are used in the hookup of various electronic assemblies.

Custodians:
Army - MI
Air Force - 80
Navy - AS

Review Activities:
Army - EL, MU
Air Force - 11
Other - IS

Preparing Activity:
Army - MI

Project No. 6145-0628

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T22:16Z
Check the source to verify that this is the current version before use.